

Bacterial Dimethylglycine and Sarcosine Catabolism: A Two-Step N-Demethylation Module Built from Interchangeable Flavoenzyme Parts

A commissioned review-style synthesis for a molecular-biology audience.

1. Executive Summary

Bacterial dimethylglycine (DMG) and sarcosine catabolism is a discrete, reusable **two-step N-demethylation module** that sequentially removes methyl groups from the tertiary amine DMG to yield the secondary amine sarcosine, and then from sarcosine to yield the primary amino acid glycine. Both reactions are carried out by covalent- or noncovalent-flavin oxidoreductases belonging to a single amine-oxidase superfamily. Each demethylation proceeds through a common chemical logic: oxidation of the amine C–H bond generates a labile iminium (an "activated formaldehyde" equivalent), whose one-carbon unit is then either captured onto tetrahydrofolate (THF) as 5,10-methylene-THF or released as free formaldehyde. This shared chemistry unifies enzymes that otherwise differ dramatically in quaternary structure, cofactor complement, electron-acceptor logic, and folate dependence.

The module's defining feature is **conserved core chemistry executed by interchangeable parts**. The first step (DMG → sarcosine) can be performed by a membrane-associated *Pseudomonas*-type DgcAB flavin/iron–sulfur dehydrogenase, by a bifunctional monomeric dimethylglycine oxidase (DMGO, *Arthrobacter*), or by an unusual single-chain NAD⁺-linked dehydrogenase (*Chromohalobacter salexigens* csal_0990/DdhC) that can act on both DMG and sarcosine. The second step (sarcosine → glycine) can be performed by a THF-coupled diflavin **heterotetrameric sarcosine oxidase (TSOX / SoxBDAG)** or by a minimalist **monomeric sarcosine oxidase (MSOX)** carrying a single covalent FAD. The order of the two steps is chemically enforced and non-reversible: sarcosine oxidases physically cannot oxidize the tertiary amine DMG, so the pathway must run DMG → sarcosine → glycine.

Biologically, the module functions as a **mobile, transcriptionally gated ecological cassette** for mining the abundant marine and host-associated osmolyte glycine betaine (GB) as a source of carbon, nitrogen, one-carbon units, and energy. It sits downstream of GB demethylation (gbcAB) and upstream of glycine cleavage/serine conversion, both of which lie outside the module boundary. It is disseminated by horizontal gene transfer (e.g., the 37-kbp genomic island of *Colwellia psychrerythraea* 34H) and is controlled by dedicated AraC/XylS-family regulators (GbdR responding to GB/DMG; SouR responding to sarcosine), whose wiring is rebuilt lineage-by-lineage around a conserved enzymatic core. The deeply conserved ancestral architecture appears to be the ETF-coupled, THF-dependent dehydrogenase (exemplified by mammalian DMGDH/SDH and the bacterial DgcAB-type systems); oxidase variants that use O₂ and folate-independent formaldehyde-releasing variants are best read as later elaborations and lineage-specific adaptations.

2. Definition and Biological Boundaries

2.1 What the module is

The system comprises exactly two sequential oxidative demethylations:

1. **DMG → sarcosine** (removal of one N-methyl group from a tertiary amine)
2. **Sarcosine → glycine** (removal of the last N-methyl group from a secondary amine)

Each step consumes molecular oxygen or transfers electrons to the respiratory chain, and each liberates a one-carbon unit that is disposed of either onto THF (as 5,10-methylene-THF) or as free formaldehyde. The product glycine is the terminal output of the module.

2.2 What lies outside the boundary — and what is often confused with it

Several neighboring processes are chemically and genetically adjacent and are frequently conflated with the module, but should be treated separately:

- **Upstream — glycine betaine demethylation (GB → DMG).** In *Pseudomonas aeruginosa*, GB → DMG requires a *separate* two-gene system, **gbcAB** (PA5410–PA5411). Mutants in gbcAB grow on DMG but not on GB, cleanly demarcating this as a distinct upstream step ([PMID: 17951379](#)). GB itself is a metabolic branch point: it may be catabolized or retained as an osmoprotectant, and this decision is regulatory, not enzymatic ([PMID: 23354714](#)).

- **Further upstream — choline oxidation to GB (betBA)** and carnitine/acylcarnitine catabolism that feeds into GB. These provide substrate but are not part of the module.
- **Downstream — glycine cleavage system (GCV) and serine hydroxymethyltransferase (GlyA).** Glycine is either cleaved (GCV, which also uses a THF-binding T-protein homologous to part of the module's enzymes — a source of confusion) or converted to serine. These consume the module's product but are outside it.
- **The THF-binding T-protein fold.** Because DMGO and TSOX's α -subunit share a THF-binding fold with the glycine-cleavage T-protein, YgfZ, and TrmE, homology-based annotation can blur module boundaries. The fold is a *shared structural prototype*, not evidence that these are the same pathway ([PMID: 16042597](#); [PMID: 15489424](#)).

2.3 Competing definitions

The main definitional tension in the literature is whether the "module" is defined by **enzyme architecture** or by **reaction chemistry**. Genetically oriented studies name it by operon (dgcAB + soxBDAG), whereas biochemically oriented studies group it by cofactor and one-carbon disposal (folate-coupled vs. formaldehyde-releasing; dehydrogenase vs. oxidase). This review adopts the chemistry-first definition because it is the invariant across lineages, while noting that the enzyme architecture is the interchangeable layer.

3. Mechanistic Overview

3.1 The best current model

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Glycine betaine (GB)
|   gbcAB  (UPSTREAM – outside module)
▼
Dimethylglycine (DMG) [tertiary amine]
|   STEP 1: DMG dehydrogenase / oxidase
|   (DgcAB | DMGO | csal_0990/DdhC)
|   covalent or noncovalent FAD; amine C–H oxidation → iminium
|   1-C disposal: → 5,10-CH2-THF   OR   → free HCHO
▼
Sarcosine (N-methylglycine) [secondary amine]
|   STEP 2: sarcosine oxidase
|   (TSOX/SoxBDAG heterotetramer | MSOX monomer)
|   covalent FAD; amine C–H oxidation → iminium
|   1-C disposal: → 5,10-CH2-THF   OR   → free HCHO
▼
Glycine [primary amine]
|   GCV cleavage / GlyA → serine  (DOWNSTREAM – outside module)
▼
central metabolism (C, N, 1-C units, energy)

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Each step is mechanistically a flavin-dependent amine oxidation: the FAD abstracts a hydride/electron pair from the substrate C–H adjacent to nitrogen, producing a reactive iminium cation and reduced flavin. The iminium is hydrolyzed — either after nucleophilic attack by the THF N10 (giving 5,10-methylene-THF and avoiding toxic free formaldehyde) or by water (giving free formaldehyde). The reduced flavin is then re-oxidized, and here the two "electron-acceptor logics" diverge (Section 3.3). A detailed mechanistic framework for how flavoproteins abstract the amine C–H — via radical versus polar routes — is reviewed for exactly this enzyme set (trimethylamine dehydrogenase, MSOX, DMGO) by [PMID: 15565251](#).

3.2 Obligatory, conditional, and accessory steps

- **Obligatory:** Both demethylations are obligatory to convert DMG all the way to glycine, and they must occur in the order DMG → sarcosine → glycine (Section 6).
- **Conditional:** Folate coupling. In *Arthrobacter* DMGO and TSOX, THF is a cosubstrate and the one-carbon unit is captured as 5,10-CH₂-THF. In MSOX and in the entire *Paracoccus denitrificans* route, folate is *not* used and formaldehyde is released, which is conditional on a formaldehyde-detoxification response being present.

- **Accessory:** The transcriptional regulators (GbdR, SouR) and transporters that supply substrate and gate expression are essential *in vivo* for growth on GB/sarcosine but are not part of the catalytic chemistry.

3.3 Two electron-acceptor logics

A central mechanistic axis distinguishes **dehydrogenases** from **oxidases**:

Property	Dehydrogenase logic	Oxidase logic
Terminal electron acceptor	Electron-transfer flavoprotein (ETF) → ETF-ubiquinone oxidoreductase → respiratory chain	Molecular O ₂
Byproduct	Reducing equivalents to the chain (energy conserving)	H ₂ O ₂
Exemplars	Mammalian DMGDH/SDH; bacterial DgcAB-type flavin/Fe–S systems; csal_0990 (NAD ⁺ -linked)	DMGO, TSOX, MSOX
Folate coupling	Yes (both mammalian dehydrogenases produce 5,10-CH ₂ -THF)	DMGO & TSOX yes; MSOX no

Mammalian DMGDH and sarcosine dehydrogenase (SDH) reoxidize their covalent FAD via ETF, which passes electrons to the main respiratory chain through ETF-ubiquinone oxidoreductase ([PMID: 18937046](#)); both use THF and convert it to N5,10-methylene-THF ([PMID: 6180732](#)). This is the paradigm for the bacterial DgcAB-type dehydrogenase, which respire the electrons rather than reducing O₂.

4. Major Molecular Players and Active Assemblies

4.1 Step 1 enzymes (DMG → sarcosine)

(a) ***Pseudomonas*-type DgcAB.** In *P. aeruginosa*, ¹³C-NMR combined with transposon mutagenesis showed that **dgcAB (PA5398–PA5399)** mutants are blocked specifically at DMG → sarcosine ([PMID: 17951379](#)). DgcAB is a membrane-associated flavin/iron–sulfur, ETF-linked dehydrogenase system — the "dehydrogenase architecture" of the module.

(b) Bifunctional dimethylglycine oxidase (DMGO), *Arthrobacter globiformis*. Crystal structures reveal a two-part enzyme: an N-terminal **covalent-FAD domain** that oxidizes DMG to a labile iminium, fused to a C-terminal **three-domain THF-binding ring** homologous to the glycine-cleavage T-protein (PMID: 12912903). The two active sites are ~40 Å apart and linked by a ~10,000 Å³ internal cavity; the THF funnel activates the folate N10 and sequesters the iminium so that toxic free formaldehyde is not released (PMID: 12912903). DMGO is the prototype of a widespread THF-binding fold (PMID: 16042597). Mechanistic mutagenesis identifies Tyr-259 as the residue that deprotonates the substrate amine prior to FAD reduction, with His-225 tuning its pK_a (PMID: 16964976). Genetically, the *Arthrobacter* dm_g gene encodes a novel dimethylglycine oxidase related to eukaryotic DMG dehydrogenase and containing nucleotide-binding, flavinylation, and folate-binding motifs (PMID: 11422368).

(c) Single-chain NAD⁺-linked dimethylglycine dehydrogenase (csal_0990 / DdhC), *Chromohalobacter salexigens*. In *C. salexigens* DSM 3043, csal_0990–0993 are required for DMG → sarcosine. csal_0990 is an "unusual" 79-kDa **monomer with noncovalently bound FAD** that uniquely uses **both DMG and sarcosine** as substrates, shows dual coenzyme specificity preferring **NAD⁺ over NADP⁺** (optimum pH 7.0, 60 °C), and is therefore a true NAD-linked dehydrogenase — distinct from the covalent-FAD, folate-coupled oxidase DMGO (PMID: 32631860). This is the review's "DdhC" single-chain exemplar and a rare enzyme that can execute both module steps.

4.2 Step 2 enzymes (sarcosine → glycine)

(a) Heterotetrameric sarcosine oxidase (TSOX / SoxBDAG). In *P. aeruginosa*, soxBDAG mutants are blocked at sarcosine → glycine (PMID: 17951379). The *Pseudomonas maltophilia* enzyme was solved at 1.85 Å: it is an αβγδ **diflavin metalloenzyme containing three coenzymes — FAD, FMN, and NAD⁺** (PMID: 16820168). FAD (β-subunit) is the site of sarcosine oxidation; a covalent FMN at the α/β interface reoxidizes it; and in the presence of THF the oxidation is coupled to formation of 5,10-methylene-THF (PMID: 16820168). The two active sites are ~35 Å apart, connected by a ~10,000 Å³ cavity — the same channeling logic as DMGO. The covalent FMN link (β-His173) is essential for activity and subunit assembly (PMID: 11330998). The sox operon is arranged glyA–soxBDAG, physically linking sarcosine oxidation to serine hydroxymethyltransferase and folate one-carbon metabolism (PMID: 7543100). Kinetic and isotope studies of the *Arthrobacter* 1-IN TSOX further show C–H bond cleavage proceeds by ground-state quantum tunneling driven by protein dynamics (PMID: 10684595).

(b) Monomeric sarcosine oxidase (MSOX). MSOX (~44 kDa, *Bacillus/Arthrobacter*) is among the simplest members of the covalent-flavin amine-oxidase family, carrying a single 8 α -S-cysteinyl FAD, requiring no folate, and releasing free formaldehyde (PMID: 10368302). Steady-state kinetics show it oxidizes secondary amino acids (sarcosine $k_{cat} \approx 7030 \text{ min}^{-1}$; N-methyl-L-alanine $\approx 8700 \text{ min}^{-1}$; N-ethylglycine; L-proline) but **not** the tertiary amine N,N-dimethylglycine (PMID: 10913293).

4.3 Family relationships

Sequence analysis shows the module's enzymes form one superfamily: the TSOX β -subunit (covalent-FAD, sarcosine oxidation) is homologous to the N-terminal half of DMG dehydrogenase and to monomeric sarcosine oxidases, while the TSOX α -subunit C-terminus and DMG dehydrogenase C-terminus share the T-protein-like THF-binding domain (PMID: 7543100). The family extends to N-methyltryptophan oxidase and pipecolate oxidase, and more distantly to mammalian SDH/DMGDH. All share the mechanistic theme of flavin-mediated amine C–H oxidation (PMID: 15565251).

4.4 Comparative summary table

Feature	DgcAB (<i>Pseudomonas</i>)	DMGO (<i>Arthrobacter</i>)	csal_0990/DdhC (<i>Chromohalobacter</i>)	TSOX/ SoxB DAG	MSOX
Step	DMG→Sar	DMG→Sar	DMG→Sar (& Sar→Gly)	Sar→Gly	Sar→Gly
Quaternary structure	Membrane flavin/Fe–S	Bifunctional monomer	Monomer, 79 kDa	Heterotetramer $\alpha\beta\gamma\delta$	Monomer ~44 kDa
Flavin	FAD	Covalent FAD	Noncovalent FAD	FAD + covalent FMN (+NAD ⁺)	Covalent 8 α -S-Cys FAD
Electron acceptor	ETF/respiratory chain	O ₂	NAD ⁺	O ₂	O ₂
Folate coupling	Yes (dehydrogenase paradigm)	Yes (5,10-CH ₂ -THF)	Not required	Yes (5,10-CH ₂ -THF)	No (free HCHO)
One-carbon output	5,10-CH ₂ -THF	5,10-CH ₂ -THF	(dehydrogenase)	5,10-CH ₂ -THF	Free formaldehyde

5. Evolutionary and Cell-Biological Variation

5.1 Variation across lineages

The module is found across Gammaproteobacteria (*Pseudomonas*, *Chromohalobacter*, *Colwellia*), Actinobacteria (*Arthrobacter*, *Corynebacterium*), Alphaproteobacteria (*Paracoccus*), and Betaproteobacteria (*Burkholderia*). The invariant is the two-step chemistry; the variables are enzyme architecture, folate coupling, electron acceptor, and regulatory wiring.

- **Architecture swaps.** *Pseudomonas* uses DgcAB + heterotetrameric SoxBDAG; *Arthrobacter* uses bifunctional DMGO + heterotetrameric SoxBDAG; *Chromohalobacter* uses a single-chain NAD-linked dehydrogenase; monomeric MSOX replaces the heterotetramer in other taxa. These are genuinely interchangeable parts achieving the same net transformation.
- **Folate coupling vs. formaldehyde release.** In *Paracoccus denitrificans*, the oxidative demethylations of GB, DMG, and sarcosine are all **sources of formaldehyde** — up to three, two, and one equivalents respectively — with no folate capture ([PMID: 38501746](#)). This mandates a dedicated formaldehyde-detoxification response controlled by the **FlhSR** histidine-kinase/response-regulator pair; flhSR mutants fail to grow on choline ([PMID: 38501746](#)).

5.2 Physiological states and ecology

The module is deployed as an **ecological cassette for osmolyte mining**. GB is one of the most abundant compatible solutes in marine and host environments, and the decision to catabolize versus retain it is state-dependent ([PMID: 23354714](#)). In the psychrophilic marine bacterium *Colwellia psychrerythraea* 34H, a **duplicated 37-kbp high-GC genomic island** carries a heterotetrameric sarcosine oxidase operon adjacent to genes for serial GB → DMG → sarcosine → glycine demethylation; this was identified as an **inter-order horizontal gene transfer event** enabling compatible-solute catabolism, and the organism grows on sarcosine as sole C and N source ([PMID: 23674353](#)). In *Methylobacterium*, DMG and sarcosine catabolism is integrated into the broader GB-utilization network and contributes to osmotic stress protection ([PMID: 38934615](#)).

5.3 Regulatory rewiring around a conserved core

- In *P. aeruginosa*, **GbdR** (AraC/XylS family, GATase1-containing) induces gbcAB and dgcAB in response to GB/DMG, while **SouR (PA4184)**, a sarcosine-specific AraC-family regulator, is

required for appreciable growth on sarcosine as a C and N source and induces the sox operon (PMID: 17951379; PMID: 26503852).

- gbdR itself is transcribed from a σ^{54} **promoter** under NtrC (nitrogen), CbrB (carbon), IHF, and BetI control, tying module expression to global carbon/nitrogen status (PMID: 28791946).
- In the distantly related *Burkholderia thailandensis*, the same catabolic logic is rebuilt with distinct gene arrangement and multiple GATase1-containing AraC regulators (gbdR1, gbdR2, and a souR ortholog), demonstrating **regulatory rewiring around a conserved enzymatic core** (PMID: 27381916).

5.4 Cross-domain conservation

Mammalian mitochondria run a homologous two-step demethylation (DMGDH and SDH) with covalent FAD, ETF coupling, and THF cofactor use (PMID: 18937046; PMID: 6180732). The covalent FAD of these enzymes is even a shared autoantigen (anti-M7) with bacterial sarcosine dehydrogenase, underscoring deep structural conservation (PMID: 9528896). This suggests the **ETF-coupled, THF-dependent dehydrogenase is the deeply conserved ancestral form**, from which O₂-using oxidases and folate-independent formaldehyde-releasing variants are lineage-specific elaborations.

6. Constraints, Dependencies, and Failure Modes

6.1 The order is chemically enforced

The most important constraint is **substrate specificity dictating reaction order**. Sarcosine oxidases act only on secondary amino acids; MSOX oxidizes sarcosine, N-methyl-L-alanine, N-ethylglycine, and L-proline, but **N,N-dimethylglycine, a tertiary amine, is not a substrate** (PMID: 10913293). Because DMG must first be demethylated to the secondary amine sarcosine before any sarcosine oxidase can act, the two steps are strictly sequential and non-interchangeable. A reversed order (sarcosine oxidase acting first on DMG) is chemically excluded. The one exception is a broadened-specificity enzyme like csal_0990/DdhC, which can perform both steps because it accepts both DMG and sarcosine (PMID: 32631860).

6.2 Channeling and the formaldehyde hazard

The iminium intermediate and free formaldehyde are toxic. Folate-coupled enzymes solve this by **substrate channeling**: DMGO and TSOX each place their two active sites ~35–40 Å apart, connected by a ~10,000 Å³ internal cavity that sequesters the iminium and delivers its carbon to THF N10 (PMID: 12912903; PMID: 16820168). Enzymes that skip folate (MSOX; the whole *Paracoccus* route) release formaldehyde and therefore **depend on a separate detoxification system** as a mandatory accessory. The *Paracoccus* FlhSR system is the clearest example: without it, formaldehyde accumulates and growth on choline fails (PMID: 38501746).

6.3 Cofactor and assembly dependencies

- The **covalent FMN link** in TSOX (β-His173) is essential for catalysis and for assembly of the β and δ subunits; His173Asn mutants form labile, inactive complexes that shed subunits, and NAD⁺ binding to the α-subunit is required for enzyme expression, indicating a cofactor-dependent folding checkpoint (PMID: 11330998).
- Active-site base chemistry: in DMGO, loss of Tyr-259 slows FAD reduction ~1500-fold, showing the reductive half-reaction depends on precise proton abstraction (PMID: 16964976).

6.4 Failure modes

- **Loss of formaldehyde detoxification** in a folate-independent lineage is lethal on GB-derived substrates (PMID: 38501746).
- **Loss of the sarcosine-specific regulator SouR** abolishes appreciable growth on sarcosine despite intact enzymes (PMID: 26503852).
- **Human DMGDH deficiency** (e.g., variant H109R) illustrates the physiological cost of losing the first step in the homologous mammalian pathway (PMID: 18937046).

7. Controversies and Open Questions

Strongly supported claims. The gene→step assignments (dgcAB → step 1; soxBDAG → step 2) are supported by combined ¹³C-NMR and mutagenesis (PMID: 17951379). The dual enzyme architectures, the diflavin/THF-channeling structures, the strict substrate-enforced order, and the two electron-acceptor logics are all backed by crystallography, kinetics, and steady-state substrate profiling.

Areas of genuine variation and possible confusion. - *Folate coupling is not universal.* The textbook "oxidative demethylation → 5,10-CH₂-THF" picture holds for DMGO and TSOX but fails for MSOX and for the entire *Paracoccus* route, which release formaldehyde (PMID: 38501746; PMID: 10368302). Generalizing folate coupling to "all bacterial sarcosine/DMG demethylation" would be an overreach. - *Mixing organisms.* Much mechanistic detail comes from mammalian DMGDH/SDH and from *Arthrobacter/Corynebacterium* oxidases; extrapolating these to the membrane-bound *Pseudomonas* DgcAB dehydrogenase (whose structure and Fe-S/ETF coupling are inferred rather than crystallographically solved) is reasonable but not directly demonstrated. - *Regulatory architecture is not conserved.* The GbdR/SouR logic of *Pseudomonas* is rebuilt with different gene arrangements and regulator counts in *Burkholderia*, so regulatory conclusions should not be transferred wholesale between lineages (PMID: 27381916).

Most important open questions. 1. What is the structure and detailed electron-transfer path of the *Pseudomonas* DgcAB flavin/Fe-S dehydrogenase? It is defined genetically but not structurally. 2. How widespread is the folate-independent (formaldehyde-releasing) mode across bacterial lineages, and does it always co-occur with a dedicated formaldehyde-detox regulon like FlhSR? 3. In organisms with a dual-substrate single-chain enzyme (csal_0990/DdhC), is the second demethylation still performed by a separate sarcosine oxidase in vivo, or can one enzyme carry both steps physiologically? 4. What determines the evolutionary choice between energy-conserving dehydrogenase (ETF/respiratory) and O₂-consuming oxidase modes — oxygen availability, energy economy, or regulatory constraints? 5. Which extant family member best represents the ancestral role? The ETF-coupled, THF-dependent dehydrogenase is the leading candidate, but this rests on conservation arguments rather than ancestral reconstruction.

8. Limitations and Knowledge Gaps

This review is a literature synthesis of 35 papers, not a primary experimental study; no new sequences or structures were generated. Key limitations:

- **DgcAB is under-characterized structurally.** Its assignment rests on genetics and homology; its cofactor complement and ETF coupling are inferred from the mammalian dehydrogenase paradigm.
- **Uneven organismal sampling.** Mechanistic depth is concentrated in a handful of model organisms (*Arthrobacter*, *Corynebacterium*, *Pseudomonas maltophilia*, *P. aeruginosa*,

Chromohalobacter, *Paracoccus*). Environmental diversity (uncultured marine taxa carrying HGT islands) is largely unexplored at the biochemical level.

- **Ancestral-state claims are inferential.** The proposal that the ETF/THF dehydrogenase is ancestral is based on cross-domain conservation, not on phylogenetic ancestral reconstruction.
- **In vivo flux partitioning** between folate capture and formaldehyde release under different growth conditions is not quantified for most organisms.

9. Proposed Follow-up Experiments / Actions

1. **Structural determination of DgcAB.** Solve a cryo-EM or crystal structure of a *Pseudomonas*-type DgcAB complex to confirm the flavin/Fe-S composition and the ETF-docking interface, closing the biggest structural gap in the module.
 2. **Phylogenomic census.** Systematically survey sequenced genomes/metagenomes for co-occurrence of (dgcAB-like OR single-chain DMGDH) with (soxBDAG OR MSOX) and with folate-capture vs. formaldehyde-detox genes (e.g., FlhSR, glutathione-dependent formaldehyde dehydrogenase), to map how often each modular variant occurs and in which ecological niches.
 3. **Ancestral sequence reconstruction.** Build a phylogeny of the covalent-FAD amine-oxidase superfamily (MSOX, TSOX- β , DMGDH/SDH N-terminal domain, N-methyltryptophan oxidase) and resurrect the inferred ancestor to test whether it is a broad-specificity dehydrogenase or oxidase.
 4. **In vivo one-carbon fate tracing.** Use ^{13}C -labeled DMG/sarcosine plus isotope-resolved metabolomics in folate-coupled (*Arthrobacter*) vs. formaldehyde-releasing (*Paracoccus*) strains to quantify the branch between 5,10-CH₂-THF and free formaldehyde under varying O₂ and folate status.
 5. **Dual-substrate enzyme physiology.** In *C. salexigens*, test whether csal_0990/DdhC alone can support growth on sarcosine when the sarcosine oxidase is deleted, resolving whether a single enzyme can carry both module steps in vivo.
 6. **Regulator swap experiments.** Test cross-complementation of *Pseudomonas* GbdR/SouR with *Burkholderia* orthologs to quantify how much regulatory rewiring the conserved enzymatic core tolerates.
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10. Key References

- Wargo MJ et al. *Identification of two gene clusters and a transcriptional regulator required for Pseudomonas aeruginosa glycine betaine catabolism*. PMID: 17951379 — assigns dgcAB (step 1), soxBDAG (step 2), and gbcAB (upstream boundary).
- Leys D, Basran J, Scrutton NS. *Channelling and formation of 'active' formaldehyde in dimethylglycine oxidase*. PMID: 12912903 — DMGO domain architecture and iminium channeling.
- *Crystal structure of DMGO provides a prototype for a new tetrahydrofolate-binding fold*. PMID: 16042597 — THF-binding fold shared with T-protein/YgfZ/TrmE.
- *Heterotetrameric sarcosine oxidase: structure of a diflavin metalloenzyme at 1.85 Å*. PMID: 16820168 — TSOX three-coenzyme architecture and THF coupling.
- *Organization of the multiple coenzymes and subunits and role of the covalent flavin link in TSOX*. PMID: 11330998 — essentiality of the covalent FMN link and NAD⁺-dependent assembly.
- *Kinetic studies of C–H bond breakage by heterotetrameric sarcosine oxidase (Arthrobacter 1-IN)*. PMID: 10684595 — quantum-tunneling C–H cleavage in TSOX.
- *Monomeric sarcosine oxidase: structure of a covalently flavinylated amine oxidizing enzyme*. PMID: 10368302 — minimalist single-covalent-FAD architecture, folate-independent.
- *Monomeric sarcosine oxidase: 2. Kinetic studies*. PMID: 10913293 — DMG is not an MSOX substrate; enforces reaction order.
- *Sequence analysis of sarcosine oxidase and nearby genes reveals homologies with folate one-carbon metabolism*. PMID: 7543100 — glyA–soxBDAG operon and superfamily homologies.
- *Organization of DMG and sarcosine degradation genes in Arthrobacter spp.* PMID: 11422368 — dmg/soxBDAG gene organization and folate-dependent pathway.
- Yang et al. *Role of csal genes in Chromohalobacter salexigens*. PMID: 32631860 — single-chain NAD⁺-linked dual-substrate DMG dehydrogenase (DdhC).
- Parekh, Tsai & Spiro. *Choline degradation in Paracoccus denitrificans*. PMID: 38501746 — folate-independent formaldehyde release and FlhSR detox control.
- *An inter-order horizontal gene transfer event ... Colwellia psychrerythraea 34H*. PMID: 23674353 — module as an HGT genomic island.
- *Enhanced catabolism of glycine betaine and derivatives in Methylobacteria*. PMID: 38934615 — DMG/sarcosine utilization in osmotic-stress protection.

- *Sarcosine catabolism in Pseudomonas aeruginosa is transcriptionally regulated by SouR.* PMID: 26503852 — sarcosine-specific regulation.
- *Homeostasis and catabolism of choline and glycine betaine.* PMID: 23354714 — GB as branch point.
- *P. aeruginosa gbdR is transcribed from a σ 54-dependent promoter.* PMID: 28791946 — global C/N control of module expression.
- *Choline catabolism in Burkholderia thailandensis.* PMID: 27381916 — regulatory rewiring around a conserved core.
- *Molecular basis of dimethylglycine dehydrogenase deficiency (H109R).* PMID: 18937046 — ETF/respiratory dehydrogenase logic.
- *The effect of tetrahydrofolate on the reduction of ETF by sarcosine and dimethylglycine dehydrogenases.* PMID: 6180732 — THF cosubstrate of both dehydrogenases.
- *Mechanism of FAD reduction ... Arthrobacter globiformis DMGO (His-225/Tyr-259).* PMID: 16964976 — active-site base chemistry.
- *Chemical aspects of amine oxidation by flavoprotein enzymes.* PMID: 15565251 — mechanistic framework for the superfamily.
- *Crystal structure of the YgfZ protein from E. coli.* PMID: 15489424 — related THF-binding fold, boundary caution.
- *Anti-mitochondrial antibodies ... covalently bound FAD.* PMID: 9528896 — shared covalent-FAD antigen across bacterial/mammalian enzymes.

Prepared as a commissioned review synthesis. Claims are anchored to the cited primary literature; uncertainty and lineage-specific variation are flagged throughout, and no single organism, cell type, or assay is treated as representative of all biology.
